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Asset management device with automatic configuration detection

Abstract

An example device includes: an enclosure; a plurality of drawer units supported in the enclosure, each drawer unit movable between an accessible position and a secure position and comprising: at least one divider configured to divide an interior of the drawer unit into a plurality of compartments; a selectively securable compartment door for each of the compartments; detection elements configured to monitor the connection points; and a control system configured to: determine, for each drawer unit, a compartment configuration; and in response to an access request: identify a designated drawer unit and a designated compartment based on the determined compartment configuration for the designated drawer unit; and release the compartment door for the designated compartment to allow access to the designated compartment.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority from U.S. Application No. 63/441,647 filed on Jan. 27, 2023, the contents of which are incorporated herein by reference.

FIELD

(1) The specification relates generally to asset management consoles, and more particularly to an asset management console with automatic configuration detection.

BACKGROUND

(2) Electronic inventory dispensing solutions for asset management may be utilized in a variety of sectors. For example, tool cabinets may store and track tools, while pharmacy applications may store and track usage and inventory levels of drugs.

SUMMARY

(3) According to an aspect of the present specification an example device includes: an enclosure; a plurality of drawer units supported in the enclosure, each drawer unit configured to move between an accessible position and a secure position relative to the enclosure, each drawer unit comprising: at least one divider configured to divide an interior of the drawer unit into a plurality of compartments; a selectively securable compartment door for each of the compartments; a plurality of detection elements configured to monitor a plurality of connection points of the drawer unit to detect the plurality of compartments; a control system configured to: determine, for each drawer unit, a compartment configuration based on detection of the compartments at the connection points by the detection elements; and in response to an access request: identify a designated drawer unit of the plurality of drawer units; identify a designated compartment based on the determined compartment configuration for the designated drawer unit; and release the compartment door for the designated compartment to allow access to the designated compartment.

(4) According to another aspect of the present specification, an example method includes: detecting, at a detection element of a drawer unit of a plurality of drawer units of the asset management device, a connection element; determining, based on the detecting, a compartment configuration of compartments in the drawer unit; receiving an access request, and in response to the access request: identifying a designated drawer unit and a designated compartment based on the compartment configuration for the designated drawer unit; and releasing a compartment door for the designated compartment to allow access to the designated compartment.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Implementations are described with reference to the following figures, in which:

(2) FIG. 1 depicts an example asset management device with automatic configuration detection in accordance with the present disclosure.

(3) FIG. 2 depicts the asset management device of FIG. 1 with a drawer unit in an accessible position.

(4) FIG. 3 depicts an example drawer unit of the asset management device of FIG. 1.

(5) FIGS. 4A-C depict example configurations of drawer units.

(6) FIG. 5 depicts a block diagram of certain internal components of the asset management device of FIG. 1.

(7) FIG. 6 depicts a flowchart of an example method of operating an asset management device with automatic configuration detection.

DETAILED DESCRIPTION

(8) Electronic inventory dispensing solutions may offer security measures and control access by selectively providing access based on each access request. However, existing electronic inventory dispensing solutions are typically static in order to allow predictability for the automatic access control.

(9) In the present example, an asset management device includes configurable compartments with embedded detection elements to allow for automatic detection of compartment configuration for further flexibility of the asset management device.

(10) FIG. 1 depicts an example asset management device **100** in accordance with the present disclosure. The asset management device **100** (also referred to herein as simply the device **100**) is generally configured to house assets or items which may be dispensed to users. For example, the device **100** may be a type of vending machine or the like to dispense items upon sale to the user, an organizational cabinet, console, or the like to store different types of equipment (e.g., including hardware, tools, medical devices, etc.), medications (e.g., including tablets, capsules, etc.), or similar. For example, the device **100** may be deployed in a facility such as a hospital, a warehouse, an office building, or other the like.

(11) The device **100** includes an enclosure **104** in which the items are housed. The enclosure **104** supports a plurality of drawer units **108**, of which four example drawer units **108-1**, **108-2**, **108-3**, and **108-4** (referred to herein generically as a drawer unit **108** and collectively as drawer units **108**; this nomenclature is also used elsewhere herein) are depicted. In other examples, the enclosure **104** may support more or fewer drawer units **108**. The enclosure **104** may include a suitable frame, and may further include wheels, motors, navigational units, batteries, and the like to support the operation and optional mobility of the device **100**.

(12) Each drawer unit **108** is configured to move between a secure position, as depicted in FIG. 1, and an accessible position, in which the interior of the drawer unit **108** is accessible by a user. For example, the drawer units **108** may slide or extend from the enclosure **104** using rails, slides, runners, belts, or the like. In some examples, the device **100** may include one or more springs, pistons, biasing mechanisms or the like to automatically move the drawer units **108**, while in other examples, each drawer unit **108** may include a handle or pull to allow a user to manually move the drawer unit **108**. In some examples, each drawer unit **108** may include a handle with lights to guide users. For example, around the drawer handle, a light-emitting diode (LED) bar may be installed. The LED bar may be configured to change color or to blink. For example, when the drawer unit is not closed properly, the LED bar may be controlled to periodically flash a red light as an alert.

(13) In still further examples, the device **100** may be configured to automatically move the drawer units **108** to the accessible position based on an access request (as will be described further herein), while expecting a user to manually return the drawer units **108** to the secure position.

(14) FIG. 2 depicts the device **100** with the drawer unit **108-2** in the accessible position, extended from the enclosure **104**. The drawer unit **108-2** is divided into a plurality of compartments **200**. Each compartment **200** includes a compartment door **204** which is selectively securable using any suitable locking mechanism, including one or more latches, pins, slides, magnets, springs, and the like.

(15) When the locking mechanism of a given compartment door **204** is released, the compartment doors **204** are similarly capable of moving between a closed position and an open position, for example by a hinge mechanism, a tambour door or roller mechanism, a sliding mechanism, or similar. Preferably, the compartment doors **204** may be configured such that the compartment doors

204 are biased to the closed position when the drawer unit **108** is returned from the accessible position to the secured position. In the present example, one of the compartment doors **204** is shown in the open position, while the other compartment doors **204** are in the closed position. In some examples, the compartment doors **204** may include one or more springs, pistons, biasing mechanisms or the like to automatically move between the open and closed positions, while in other examples, each compartment door **204** may include a handle or pull to allow a user to manually move the compartment door **204** between the open and closed positions. In still further examples, the compartment door **204** may be configured to be biased or automatically moved to the open position, while expecting a user to manually return the compartment door **204** to the closed position.

(16) FIG. 3 depicts a top view of a drawer unit **108** with the compartment doors **204** removed. In particular, the compartments **200** are defined in the drawer unit **108** by a plurality of dividers **300**, including six vertical dividers **300-1** and four horizontal divider **300-2**, resulting twenty-eight (4×7) compartments. The dividers **300** are configured to divide the interior of the drawer unit **108** into the compartments **200**. In some examples, both the vertical dividers **300-1** and the horizontal dividers **300-2** may be configurable within the drawer unit **108**. In other examples, one set of dividers (e.g., the vertical dividers **300-1**) may be fixed, while the other set of dividers (e.g., the horizontal dividers **300-2**) may be configurable.

(17) In the present example, connection points **304** may be defined on the horizontal dividers **300-2** to receive a corresponding latch of a compartment door for the resulting compartment.

(18) The drawer unit **108** further includes at least one detection element **308** configured to monitor the connection points of the drawer unit **108**. In the present example, the detection elements **308** may be integrated into the connection points **304**, and hence the drawer unit **108** includes as many detection elements **308** as connection points **304**. For example, the detection elements **308** may be a socket or the like to which a corresponding connection element **312** is configured to connect when the divider **300** is connected to the connection point **304**. In the present example, the connection element **312** may be the latch on the compartment door **204**. That is, the detection element **308** and the corresponding connection element **312** may complete or form an electronic circuit to allow the detection element **308** to monitor and identify the connection to the connection point **304**. Further, the detection element **308** may be configured to identify the compartment door **204** connected at the connection point **304**, for example by an identifier, to infer the compartment configuration.

(19) In other examples, other types of detection elements **308** are also contemplated. For example, rather than monitoring connection points **304** for latches of the corresponding compartment doors **204**, the connection points **304** may be to receive the dividers **300**. In such examples, the dividers **300** may have a predefined configuration, for example extending across an entire length or width of the drawer unit **108**, or having a predefined shape. In other examples, the configuration or shape of the dividers **300** may be inferred by the detection element **308**. For example, each detection element **308** may have two corresponding sockets, located at each end of the vertical or horizontal divider to complete an electronic circuit via the divider **300**. In other examples, the dividers **300** may encode self-identifying information which may be detected by the detection elements **308** (e.g., via communication of a signal through the detection element **308** to the connection element **312**) to identify the configuration of the dividers **300**.

(20) In still further examples, the detection elements **308** may be configured to monitor the connection points **304** in a one-to-many relationship. For example, the detection elements **308** may include one or more cameras (e.g., two cameras for stereoscopic functions) configured to capture an image of the drawer unit **108** when in the accessible position and apply image analysis of the drawer unit **108** to identify and monitor the connection points of the drawer unit **108**. For example, the camera may expect a matrix (i.e., having a length and width) of compartments and may recognize the configuration based on the expected matrix.

(21) In the present example, the detection elements **308** are interconnected with a drawer unit controller **316**. For example, the drawer unit controller **316** may include a central processing unit (CPU), a microcontroller, a microprocessor, a processing core, a field-programmable gate array (FPGA), or similar. The drawer unit controller **316** may include multiple cooperating processors. The drawer unit controller **316** may cooperate with a memory (not shown), including a suitable combination of volatile and/or non-volatile memory, some or all of which may be integrated with the drawer unit controller **316**. In particular the drawer unit controller **316** may execute computer-readable instructions stored on the memory to determine a compartment configuration of the compartments **200** based on the dividers **300** detected by the detection elements **308**. In some examples, the drawer unit controller **316** may further be interconnected with the compartment doors **204** to release the respective locking mechanisms to allow the compartment doors **204** to be moved from the closed position to the open position.

(22) Referring to FIGS. **4A-4C** three example configurations of compartments within a drawer unit **108** are depicted. In configuration **400** as depicted in FIG. **4A**, the detection elements **308** may monitor the connection points **304** to which the horizontal dividers **300-2** are configured to connect. That is, in this example, the connection points **304** may represent portions of the drawer unit **108** which may accept a divider **300** in order to divide the interior of the drawer unit **108** into compartments. For example, the connection points **304** may include a slot, a notch, or similar to receive and secure the divider **300** at the connection point **304**. The detection elements **308** may detect that the connection points **304** for the horizontal dividers **300-2** are employed to connect a respective horizontal divider **300-2**. In some examples, further detection elements **308** may detect that six connection points **304** for the vertical dividers **300-1** are employed to connect a respective vertical divider **300-1**, while in other examples, the vertical dividers **300-1** may be fixed at six vertical dividers **300-1**. In either case, the drawer unit controller **316** may determine, based on a predefined definitions of the dividers **300** that twenty-eight compartments **200** are defined.

(23) In the configuration depicted in FIG. **4B**, the detection elements **308** may similarly detect that the connection points **304** for the horizontal dividers **300-2** are employed to connect a respective horizontal divider **300-2**. In the present example, the horizontal dividers **300-2** myself-encode the configuration that they extend across only to the sixth of the fixed vertical dividers **300-1**. The drawer unit controller **316** may therefore determine, based on the identification of the configuration of the horizontal dividers **300-2** at the detection elements **308**, that the twenty-five compartments **200** are defined.

(24) In configuration **420** depicted in FIG. **4C**, connection points **304** (and corresponding detection elements **308**) may be defined on the fixed vertical dividers **300-1** in addition to the connection points **304** defined at an edge of the drawer unit **108**. For example, the vertical dividers **300-1** may include slots with detection elements **308** within which the horizontal dividers **300-2** are configured to be seated to secure the horizontal dividers **300-2** in place. Accordingly, the detection elements **308** may detect the span of each horizontal dividers **300-2** based on the connection points on the vertical dividers **300-1** to allow the drawer unit controller **316** to determine that thirteen compartments **200** are defined.

(25) In still further examples, other configurations or arrangements of dividers **300** are contemplated, along with corresponding connection points **304** within the drawer unit **108**.

(26) The control system can detect the configuration (such as drawer height) of the drawers vertically scanning the connected drawers and their configured address. Consequently, the system **100** can automatically recognize the configuration of the drawers such as the quantity/the size of compartments as well as the quantity/the height of the drawers.

(27) Turning now to FIG. **5**, a block diagram of certain internal components of the device **100** are depicted. The device **100** includes a control system **500** including a plurality of drawer unit controllers **316** and a device controller **504**.

(28) Each of the drawer unit controllers **316** is interconnected with a corresponding set of detection

elements **308** for the respective drawer unit **108** (for convenience, only one detection element **308** is illustrated; as noted above, each drawer unit **108** may include multiple detection elements **308**). That is, the drawer unit controllers **316** may be configured to communicate with the detection elements **308** to identify the dividers **300** employed in the respective drawer units **108** and to determine a compartment configuration for the drawer unit **108**.

(29) Each of the drawer unit controllers **316** may additionally be interconnected with a corresponding set of locking mechanisms **508** for each of the compartment doors **204** of the respective drawer unit **108** (for convenience, only one locking mechanism **508** is illustrated; in practice, each drawer unit **108** may include as many locking mechanisms **508** as compartment doors **204**). For example, the locking mechanisms **508** may include one or more latches, pins, slides, magnets, solenoids, and the like, configured to be electrically released via control from the drawer unit controller **316**. That is, the drawer unit controllers **316** are further configured to control the locking mechanisms **508** on each of the compartment doors **204** to selectively release one or more of the locking mechanisms **508** and allow the corresponding compartment door **204** to be moved to the open position to allow access to the corresponding compartment **200**.

(30) In some examples, the drawer unit controllers **316** may further be interconnected with one or more springs, biasing mechanisms, or the like, for each of the compartment doors **204** to automatically move the compartment door **204** to the open position after releasing the corresponding locking mechanism **508**. In other examples, the release mechanism for the compartment doors **204** may be integrated with the locking mechanism **508**.

(31) In the present example, the drawer unit controllers **316** are illustrated as being functionally part of the control system **500**; in practice the drawer unit controllers **316** may be physically located within each corresponding drawer unit **108**.

(32) In some examples, the drawer units **108** may further include one or more supplemental compartment features for the drawer unit **108** and/or localized to each compartment **200**. For example, when the dividers **300** have predefined options (e.g., shape of divider being vertical and horizontal dividers and fixed divider positions/connection points), the potential available compartments which may be defined is also fixed. Accordingly, the supplemental compartment features may be regionalized based on a smallest unit compartment and interconnected with the drawer unit controllers **316** to be activated by compartment based on the determined compartment configuration.

(33) For example, the configuration **400** shown in FIG. 4A may divide a drawer unit **108** into its unit compartments. Accordingly, in the configurations **410** or **420**, one or more of the unit compartments may be combined to form the respective larger compartments. The drawer unit controller **316** may determine which unit compartments are assigned to a given detected (larger) compartment in the compartment configuration.

(34) For example, the supplemental compartment features may include weight sensors to measure the weights of items stored in each compartment **200**. In some examples, the weight sensors may detect the weight difference when the compartment door **204** is in the open versus the closed position. For example, opening the compartment door **204** may apply or remove some pressure from the weight sensors based on the orientation of the compartment door **204**. The weight sensors may allow the device **100** to be used, for example to allow items to be collected when the weight in a given compartment **200** reaches a predefined threshold weight level. Thus, for example, the device **100** may act as a collection bin for recycling, scrapping, etc.

(35) In other examples, the supplemental compartment features may be heating and/or cooling devices and/or structures, such as temperature sensors, Peltier devices, insulation, moisture controls, and the like. The supplemental compartment features may further include lights, such as light-emitting diodes (LEDs), cameras and/or imaging units for analysis, and the like. The supplemental compartment features may further include devices for charging, sanitizing and/or cleaning, cutting/grinding, cables for power or communications, and the like.

(36) The drawer unit controllers **316** may further be interconnected with the device controller **504**, which may be a CPU, a microcontroller, a microprocessor, a processing core, an FPGA, multiple cooperating processors, or similar. The controller **504**, or more generally, the control system **500** (i.e., including the drawer unit controllers **316**), may be interconnected with a memory **512**, which may include a combination of volatile (e.g., Random Access Memory or RAM) and non-volatile memory (e.g., read-only memory or ROM, Electrically Erasable Programmable Read Only Memory or EEPROM, flash memory). The memory **512** stores applications, each including a plurality of computer-readable instructions executable by one of the controllers **316** or **504**. The execution of the instructions configures the device **100** to perform the actions discussed herein. In some examples, some or all of the functions described herein as being executed by the drawer unit controllers **316** may be performed by the device controller **504** or vice versa.

(37) For example, the applications stored in the memory **512** may include a control application **516**. When executed, the control application **516** may configure the device **100** to perform the automatic configuration detection, and to process access requests. The application **516** may also be implemented as a suite of distinct applications (e.g., including separate applications for the automatic drawer unit configuration detection and for the access request processing). Further, some or all of the functionality of the application **516** may be implemented as dedicated hardware components such as one or more FPGAs or application-specific integrated circuits (ASICs).

(38) The memory **512** also stores a repository **520** storing rules and data for the operations of the device **100**. For example, the repository **520** may store the determined compartment configurations for each of the drawer units **108**, item information for items stored in the compartments **200**, including identifying information, access information and location information (e.g., including a drawer unit identifier and a compartment identifier), and the like.

(39) The device **100** may further include a communications interface **524** interconnected with the control system **500**. The communications interface **524** may be configured for wireless (e.g., satellite, radio frequency, Bluetooth, Wi-Fi, or other suitable communications protocols) or wired communications and may include suitable hardware (e.g., transmitters, receivers, network interface controllers, and the like) to allow the device **100** to communicate with other computing devices. The specific components of the communications interface **524** are selected based on the types of communication links that the device **100** communicates over.

(40) The asset management device **100** can designate the other system when the device cannot offer a proper service. For example, when the device cannot provide an item requested by a user, the device can call/designate the other system which is available to provide the item. The user can access to the other device near to the device conveniently. The user can also search a proper device through the one's mobile device in advance.

(41) The device **100** may further include one or more input and/or output devices (not shown). The input devices may include one or more buttons, keypads, touch-sensitive display screen, mice, or the like for receiving input from an operator. The output devices may include one or more display screens, monitors, speakers, sound generators, vibrators, or the like for providing output or feedback to an operator.

(42) For example, the input and/or output devices may include a user interface module configured to provision augmented reality, virtual reality, or 3-dimensional content, graphical interfaces, voice-recognition and/or activation features, biometric detection systems, generative artificial intelligence features, and the like.

(43) The device **100** further includes a plurality of drawer unit release mechanisms **528**, for example, one drawer unit release mechanism **528** corresponding to each drawer unit **108**. The drawer unit release mechanisms **528** may be interconnected with the device controller **504**. That is, the device controller **504** may be configured to control the drawer unit release mechanisms **528** (e.g., based on an access request at the device **100**) to selectively release one or more drawer units **108** to allow the drawer units **108** to move from the secured position within the enclosure **104** to

the accessible position extended from the enclosure **104**. For example, the drawer unit release mechanisms **528** may include latches, magnets, solenoids, or the like to secure the drawer units **108** in the secure position until released by the device controller **504**. Additionally, the drawer unit release mechanism **528** may include one or more rails, slides, belts, or the like to automatically move the drawer unit **108** to the accessible position upon release by the device controller **504**.

(44) In particular, in operation, the device **100** is generally configured to automatically determine the compartment configuration for each of the drawer units **108**, store the compartment configuration, and process access requests for the storage or retrieval of items according to the compartment configurations.

(45) For example, turning now to FIG. **6**, the functionality implemented by the device **100** will be discussed in greater detail. FIG. **6** illustrates a method **600** of providing secure access for items. The method **600** will be discussed in conjunction with its performance by the device **100**, with reference to the components of FIGS. **1** to **5**. In other examples, the method **600** may be performed by other suitable devices or systems.

(46) At block **605**, the device **100** performs an initialization routine to determine its internal configuration, for example by determining a compartment configuration for each of the drawer units **108** in the device **100**. For example, the device controller **504** may signal each of the drawer unit controllers **316** to perform an operation to determine the compartment configuration for the respective drawer unit **108**.

(47) In response to the signal, each drawer unit controller **316** may for example, obtain signals from each of the connected detection elements **308** to determine if the monitored connection points **304** are in use (i.e., have a divider **300** connected thereto). For example, the drawer unit controller **316** may send a current to one pin of each detection element **308** and determine if a signal is received at a paired pin, indicating that the pins are electrically connected via a connection element **312** of a divider **300**. In some examples, the drawer unit controller **316** may additionally identify a configuration of the divider **300** itself, for example by sampling a return signal to extract an identifier of the divider **300**, or by inferring the configuration of the divider **300** based on the active connection points **304** as indicated by the detection elements **308**.

(48) Upon determining the compartment configuration, the drawer unit controller **316** may store the determined compartment configuration for the corresponding drawer unit **108** in the repository **520**. In some examples, the compartment configuration may further be stored with an association of each unit compartment within the drawer unit **108** to a corresponding compartment of the compartment configuration.

(49) At block **610**, the device **100** may optionally obtain a compartment assignment, indicating the assignment of items stored or contained in each of the compartments **200** identified in the compartment configuration. For example, the device **100** may identify that a compartment assignment is to be obtained based on a detected weight by weight sensors in each of the unit compartments. The device **100** may then send a request to a managing operator of the device **100** to identify the compartment assignment for the compartments in which items were detected by the weight sensors. In other examples, the compartment assignment may be proactively loaded and stored in the repository **520** by the managing operator.

(50) At block **615**, the device **100** may receive an access request, for example at an input device (e.g., including a screen, keyboard, etc., associated with the device **100**). For example, the access request may be a storage request or a retrieval request. If the access request is a storage request, the device **100** proceeds to block **620-1**.

(51) At block **620-1**, the device **100** obtains access parameters pertaining to the storage request. In some examples, the access parameters may be item based, while in other examples, the access parameters may be account and/or user based.

(52) For example, item-based parameters may include item information, such as an item identifier and/or item dimensions, or the like. For example, the device **100** may obtain an item identifier

(e.g., from a predefined list of items, based on a barcode, universal product code identifier or the like) and may determine dimensions, weight, or other parameters of the item based on the item identifier. In other examples, the device **100** may simply request and obtain the dimensions, weight, or other parameters of the item, or may determine the item parameters based on analysis of image and/or depth data obtained by an image and/or depth camera designated for item scanning.

(53) In such examples, the device **100** may process the storage request and assign a suitable compartment for storage based on the item information, as described below. In some examples, the item parameters may further include storage parameters, such as suitable storage temperatures or other conditions for storing the item.

(54) In other examples, additionally or alternatively, the access parameters may be account and/or user based. That is, certain users or accounts may be assigned to certain compartments within the device **100**, and accordingly, the device **100** may verify account credentials to identify the assigned compartment for storage based on the account parameters.

(55) At block **625-1**, the device **100** identifies, based on the access parameters, a designated drawer unit for the storage request. For example, the designated drawer unit may be selected based on a height or depth of the item and a corresponding height or depth of the designated drawer unit. In other examples, the designated drawer unit may be predefined based on the account.

(56) At block **630-1**, the device **100** identifies, based on the access parameters, a designated compartment based on the compartment configuration for the designated drawer unit. For example, the designated compartment may be selected on a length and width of the item and a corresponding length and width of the compartment, and/or heating and or cooling or other storage parameter capabilities of the compartment. In other examples, the designated compartment may be predefined based on the account.

(57) Since each of the drawer units **108** may have different compartment configurations, in some examples, blocks **625-1** and **630-1** may be performed in reverse order and/or substantially simultaneously. For example, only one of the drawer units **108** may have a compartment that is suitably sized to store the requested item, and hence the selection of the suitably sized compartment may inform the designation of the drawer unit.

(58) If the access request at block **615** is a retrieval request, the device **100** proceeds to block **620-2**. At block **620-2**, the device **100** obtains access parameters pertaining to the retrieval request. In some examples, the access parameters may be item based, while in other examples the access parameters may be account and/or user based.

(59) For example, item-based parameters may include item information, such as an item identifier of the item to be retrieved. In other examples, additionally or alternatively, the access parameters may be account and/or user-based. That is, certain users or accounts may be assigned to certain compartments within the device **100**, and accordingly, the device **100** may verify account credentials to identify the assigned compartment for storage based on the account parameters.

(60) In some examples, the access parameters may be based on both item parameters and account parameters. For example, certain items (e.g., certain medications, drugs, sensitive equipment, or the like) stored in the device **100** may be protected, and accordingly, when such items are requested, the device **100** may further request authentication of the user's credentials to verify that the user has sufficient permissions to retrieve the requested items. Other items may be dispensed or retrieved without such further authentication.

(61) At block **625-2**, the device **100** identifies, based on the access parameters, a designated drawer unit for the retrieval request. For example, the designated drawer unit may be selected based on the location of the requested item within the device **100**. In other examples, the designated drawer unit may be predefined based on the account.

(62) At block **630-2**, the device **100** identifies, based on the access parameters, a designated compartment based on the compartment configuration for the designated drawer unit. For example, the designated compartment may be selected based on the location of the requested item within the

device **100**. In other examples, the designated drawer unit may be predefined based on the account. (63) Blocks **625-2** and **630-2** may similarly be performed in reverse order and/or substantially simultaneously based on the nature of the retrieval request.

(64) At block **635**, the device **100** is configured to release the designated drawer unit identified at block **625**. For example, the controller **504** may send a release signal to the corresponding drawer unit release mechanism **528** of the designated drawer unit. The drawer unit release mechanism **528** may, in some examples, simply unlatch the designated drawer unit to allow the drawer unit to be moved from the secure position to the accessible position manually by the user. In other examples, the drawer unit release mechanism **528** may additionally move the designated drawer unit to the accessible position at block **635**.

(65) At block **640**, the device **100** is configured to release the compartment door for the designated compartment identified at block **630**. For example, to release the compartment door **204**, the controller **504** may signal the corresponding drawer unit controller **316** with an indication of the designated compartment. The drawer unit controller **316** may then send a release signal to the corresponding locking mechanism **508** for the designated compartment. The release of the locking mechanism **508** may then release the compartment door **204**. In some examples, releasing the locking mechanism **508** and the compartment door **204** may similarly include moving the compartment door from the closed position to the open position, while in other examples, the user may manually open the compartment door **204**.

(66) In some examples, release of the compartment door **204** may additionally initialize other supplemental compartment features of the compartment **200**, such as LED lighting to illuminate the interior of the compartment **200**, cameras and/or image sensors to monitor the retrieval or storage of the item in the compartment **200**, or the like.

(67) At block **645**, after completion of the access request, the compartment door **204** is returned to the closed position and the drawer unit **108** is returned to the secured position. In some examples, this may be manually performed by the user, while in other examples, the device **100** may be configured to perform the return operation automatically, for example after detecting that the access request is complete (e.g., based on analysis of one or more monitoring devices, based on an indication from the user), or after a predefined timeout, or the like. The device **100** may further update the compartment assignment stored in the repository **520** based on the access request.

(68) In other examples, after a predefined timeout period, the device **100** may send an error alert to indicate to the user and/or to the managing operator that the return operation was not successfully completed. For example, each of the compartment doors **204** and the drawer units **108** may be monitored to determine their state (i.e., open or closed, accessible or secured), for example, based on the state of their respective release mechanisms. If at least one compartment and/or drawer unit is not closed, the drawer unit controller **316** and/or the device controller **504** may control the drawer unit or the compartment to return to the closed and/or secured positions.

(69) In other examples, other implementations and embodiments are also contemplated. For example, the asset management device **100** may be deployed as one of a plurality of asset management devices within an asset management system deployed in a facility. In such examples, the asset management system may further include a server in communication with the asset management device **100**, and the compartment assignment may further be communicated to the server. In such examples, the asset management device **100** may additionally communicate with the other asset management devices, for example to communicate stock levels of certain inventory or the like. The asset management device **100** may further present the identity and location of nearby additional asset management devices **100** which may store requested inventory.

(70) In other embodiments, the user can call the asset management device to the user. The asset management device can be motorized with electric motors and batteries. The asset management device can be equipped with sensors and cameras to detect surroundings and location information. The server can control several asset management device to place at a proper location.

- (71) Further, the asset management device may include other control mechanisms, including augmented reality interfaces, voice recognition capability, and the like.
- (72) The scope of the claims should not be limited by the embodiments set forth in the above examples but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. An asset management device comprising: an enclosure; a plurality of drawer units supported in the enclosure, each drawer unit configured to move between an accessible position and a secure position relative to the enclosure, each drawer unit comprising: at least one divider configured to divide an interior of the drawer unit into a plurality of compartments; a selectively securable compartment door for each of the compartments, each compartment door including a latch; a plurality of detection elements configured to monitor a plurality of connection points of the drawer unit to detect the latch of each of the compartment doors; a control system configured to: determine, for each drawer unit, a compartment configuration based on detection of the latches at the connection points by the detection elements; and in response to an access request: identify a designated drawer unit of the plurality of drawer units; identify a designated compartment based on the determined compartment configuration for the designated drawer unit; and release the compartment door for the designated compartment to allow access to the designated compartment.
2. The asset management device of claim 1, wherein the at least one divider comprises: a set of horizontal dividers configured to extend across a length of the drawer unit; and a set of vertical dividers configured to extend across a width of the drawer unit.
3. The asset management device of claim 2, wherein one of the sets of horizontal dividers or vertical dividers is fixed.
4. The asset management device of claim 1, wherein the detection elements are integrated into the connection points.
5. The asset management device of claim 4, wherein the corresponding latch of the compartment door is configured to complete an electric circuit of the detection element.
6. The asset management device of claim 1, wherein the control system comprises a set of drawer unit controllers corresponding plurality of drawer units.
7. The asset management device of claim 6, wherein each drawer unit controller is configured to determine the compartment configuration for the respective drawer unit.
8. The asset management device of claim 1, wherein, to release the compartment door, the control system is configured to release a locking mechanism for the compartment door.
9. The asset management device of claim 1, wherein the control system is configured to identify the designated drawer unit and the designated compartment based on one or more of: item parameters from the access request and account parameters from the access request.
10. A method of operating an asset management device, the method comprising: detecting, at at least one detection element of a drawer unit of a plurality of drawer units of the asset management device, a latch of each corresponding compartment door; determining, based on the detecting of at least one of the latches, a compartment configuration of compartments in the drawer unit; receiving an access request, and in response to the access request: identifying a designated drawer unit and a designated compartment based on the compartment configuration for the designated drawer unit; and releasing the compartment door for the designated compartment to allow access to the designated compartment.
11. The method of claim 10, further comprising: determining the compartment configuration for each drawer unit of the plurality of drawer units.
12. The method of claim 10, further comprising: identifying the designated drawer unit and the designated compartment further based on access parameters from the access request.
13. The method of claim 10, wherein releasing the compartment door comprises releasing a locking

mechanism for the compartment door.

14. The method of claim 10, wherein releasing the compartment door comprises moving the compartment door to an open position.

15. The method of claim 10, further comprising releasing the designated drawer unit prior to releasing the compartment door.

16. The method of claim 15, further comprising moving the designated drawer unit to an accessible position.
